U.S. Department of Transportation
Federal Highway Administration

Eastern Federal Lands Highway Division
NEPA Categorical Exclusion Form

Project Number:	NJ ERFO FW WARI 2019-1(1)	**Date:** 11/6/2023
Project Name:	Wallkill River National Wildlife Refuge Trail Repairs	
Location:	Wallkill River National Wildlife Refuge Vernon Township, Sussex County, New Jersey	
CE Category	23 CFR 771.117(c)(23): Federally funded projects (i) That receive less than $6,000,000; or (ii) with a total estimated cost of not more than $35,000,000 and federal funding comprising less than 15% of total estimated project cost.	

See 23 CFR 771.117 for full description of CE categories, including additional requirements when applying (c)26-28.

Project Description

The Eastern Federal Lands Highway Division of the Federal Highway Administration (FHWA), in cooperation with the U.S. Fish and Wildlife Service (USFWS), is proposing to repair portions of the Liberty Loop and Dagmar Dale Nature trails within the Wallkill River National Wildlife Refuge (Refuge) located in Vernon Township, Sussex County, New Jersey. The Refuge was impacted by back-to-back heavy rain events in 2019 which led to widespread flooding throughout the region. Multiple trails within the Refuge were inundated with flood waters for several days and multiple culverts which cross under the Dagmar Dale Nature trail either failed or sustained structural damage from the flood event.

The proposed repairs on the Dagmar Dale Nature trail will involve replacing failed culverts which cross under the trail, resurfacing and regrading damaged portions of the trail surface, and installing cellular confinement system (CCS) along trail soft spots. The proposed repairs on the Liberty Loop Trail will involve deep patch repairs of the trail embankments, installation of a CCS, and turf re-establishment.

Additional trail work will include installation of temporary erosion and sediment control measures, vegetation clearing and grubbing, outfall stabilization, and other miscellaneous trail maintenance activities. All proposed work will occur within areas previously disturbed by the original construction and routine maintenance of each trail.

Environmental Commitments and Mitigation Measures

Erosion Control Measures:

- The Contractor will be required to prepare and implement a Storm Water Pollution Prevention Plan (SWPPP) and a Spill Prevention, Control, and Countermeasure Plan (SPCC) as required in the Standard Specifications (FP-14).
- Erosion and sediment control measures including sediment barriers, diversion berms, and filter bags will be used as necessary.
- Erosion and sediment control devices shall be made of 100% natural material as much as practicable and satisfy the performance standards outlined in the most current version of *The Standards for Soil Erosion and Sediment Control in New Jersey*.
- Any temporary erosion and sediment control devices made with plastic/synthetic materials must be removed and disposed of within 6-months of installation.
- Erosion and sediment control devices will be routinely inspected by qualified personnel to ensure they are functioning properly.
- Best Management Practices such as the watering of exposed soils will be implemented to ensure construction dust does not enter adjacent wetlands.

Wildlife/Wildlife Habitat

- To protect ESA-listed bat species, no trees larger than 3-inch DBH will be cut between April 1 and September 30. If trees must be cleared within this period, presence/absence surveys shall be conducted by a qualified biologist prior to tree removal.
- Vegetation clearing is not permitted from April 15 – August 15 to avoid take of species protected under the Migratory Bird Treaty Act. If vegetation must be cleared within this window, a qualified biologist must conduct migratory bird nesting surveys prior to the removal of vegetation. Nesting surveys must be conducted within 5 days of commencement of disturbance at the site. If active nests are found, the Contractor must maintain a species appropriate buffer from the nest until the young have fledged.
- In-stream work will occur during the bog turtle active season of April 1 - October 15 to avoid disturbance to hibernating bog turtles.
- To minimize the chance of injury to bog turtles and other native reptile and amphibian species, the Contractor must implement one of the following measures for sites in close proximity to wetlands:
    - Prior to the commencement of construction, a qualified biologist with experience monitoring native reptiles and amphibians will survey the proposed work area for bog turtles. If no turtles are found, the biologist will then oversee the immediate installation of a turtle exclusion barrier (e.g. straw bales, heavy duty silt fence, other approved barrier) that completely encompasses the work area; or

    - At the start of each construction day, a qualified biologist with experience monitoring native reptiles and amphibians ("herp" monitor) will visually "sweep" the work area for turtles immediately prior to any equipment entering the site. Careful attention will be taken in areas with vegetation or deep muck.
- Care shall be taken to minimize in-stream work between May 1 and July 31 to minimize impacts to non-trout fishery resources.
- Care will be taken not to disturb any wildlife species found nesting, estivating, or otherwise living in or immediately adjacent to the worksites.

General Mitigation Measures:

- All construction equipment entering the worksite must be free of any vegetation and/or vegetation parts (seeds, stems, etc.) to avoid introduction of invasive species.
- No equipment is authorized enter wetland areas adjacent to the Liberty Loop Trail.
- The Contractor will make every reasonable effort to minimize construction noise through abatement measures such as proper maintenance of construction equipment and use of modern equipment with better engine insulation and mufflers.
- Idling Vehicles – construction vehicles and equipment must adhere to the idling standards (less than 3 minutes) in N.J.A.C. 7:27-14 and 15.

Summary of Section 7 of the Endangered Species Act Compliance

The USFWS Information for Planning and Consultation (IPaC) website was used to obtain an official species list on June 1, 2023. Three listed species, one proposed species, and one candidate species were identified as having the potential to occur in the project area. USFWS prepared an Intra-Service Section 7 Biological Evaluation (BE) for the proposed project and determined that the proposed action would have *no effect* on the Monarch Butterfly, and *may affect, but is not likely to adversely affect* the Northern long-eared bat, Indiana Bat, Tricolored Bat, and Bog Turtle. Additionally, the New Jersey Ecological Field Office recommended the BE also address the Eastern Black rail (*Laterallus jamaicensis*) as they had data that suggested the species may utilize modified wetlands adjacent to the Liberty Loop Trail. In the BE, USFWS determined the proposed action *may affect, but is not likely to adversely affect* the Eastern Black rail.

Section 7 compliance will be revisited prior to project construction to ensure no changes have been made to the status of listed, proposed, or candidate species.

Summary of Section 106 of the National Historic Preservation Act Compliance

FHWA defined the project Areas of Potential Effect (APE) as the existing trail surface and trail embankments within the two project areas. Coordination with the USFWS Region 5 Historic Preservation Officer (RHPO) found that several archaeological sites have been identified within the Refuge and surrounding area, including a Late Archaic to Early Woodland site near the Dagmar Dale Nature trail APE. Additionally, the Appalachian National Scenic Trail overlaps with the Liberty Loop Trail APE for approximately 1 mile before turning onto Oil City Road near the New York-New Jersey boarder. A review of the New Jersey Department of Environmental Protection, Land Resource Protection Web Application found that the portion of the Appalachian National Scenic Trail that overlaps with the project APE is eligible for listing in the National Register of Historic Places (SHPO Inventory ID 2778).

On May 23, 2023, FHWA requested concurrence from the New Jersey State Historic Preservation Office that the proposed project would have *no adverse effect* on either historic resource as all proposed work would occur on the existing trail surface and within the existing trail prism, and would not alter the surrounding viewshed. On June 20, 2023, the Deputy State Historic Preservation Officer concurred with FHWA's finding that the project, as proposed, will not adversely affect historic properties.

Additional Agency Coordination and Compliance

FHWA requested a Natural Heritage Data Review from the New Jersey Department of Environmental Protection, Office of Natural Lands Management on June 26, 2023, to obtain a list of rare wildlife and plant species or habitat that have the potential to be in the project vicinity and may be impacted by the project. Appropriate avoidance and minimization measures have been applied to the project to avoid or minimize impacts to these species.

The Refuge will coordinate with the National Park Service regarding detours for uses of the Appalachian National Scenic Trail during project construction. AT users currently utilize the southern and western loop of the Liberty Loop Trail. During project construction, detour signs will be erected to direct AT users to use the eastern side of the Liberty Loop Trial and continue to Oil City Road.

Categorical Exclusion Recommendation:
On the basis of the environmental impact information in the statutory compliance file, with which I am familiar, I believe the project should be categorically excluded from further NEPA analysis.

KIMBERLY DIANE CAMPO-ALLEN
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Kim Campo-Allen
Environmental Protection Specialist
Eastern Federal Lands Highway Division
Federal Highway Administration

Date

Categorical Exclusion Determination:
On the basis of the environmental impact information in the statutory compliance file, with which I am familiar, I am categorically excluding the described project from further NEPA analysis. The project meets the 23 CFR 771.117(a) definition of a categorical exclusion. No unusual circumstances per 23 CFR 771.117(b) apply.

KEVIN S ROSE
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Kevin S. Rose
Environmental Team Leader
Eastern Federal Lands Highway Division
Federal Highway Administration

Date

Appendix A: Environmental Checklist

Section 7 of the Endangered Species Act:
• Are Federally-listed species potentially present in the study area? ☒Yes, see Notes[i] ☐No If Yes, effects determination: ☒No Effect ☒May Affect, Not Likely to Adversely Affect ☐May Affect, Likely to Adversely Affect
Section 106 of the National Historic Preservation Act:
• Are historic properties known to be present in the study area? ☒Yes ☐No If Yes, will they be impacted by the project? ☐Yes ☒No • Will previously undisturbed ground be disturbed? ☐Yes ☒No • Effects determination: ☐No Historic Properties Affected ☒No Adverse Effect ☐Adverse Effect, MOA Executed • Was tribal consultation completed? ☐Yes ☒No
Water and Wetlands:
• Section 404 of the Clean Water Act Are impacts to Waters of the US anticipated? ☒Yes ☐No If Yes, approval anticipated: ☒Nationwide Permit (NWP) ☒Regional General Permit ☐Individual Permit • Section 401 of the Clean Water Act ☒NWP certified by State ☐Individual Certification • Section 402 of the Clean Water Act (National Pollutant Discharge Elimination System) Will the land disturbance threshold likely be exceeded to require a permit and SWPPP? ☒Yes ☐ No • Is post-construction stormwater management review/approval anticipated? ☐Yes ☒No • ☒Project is consistent with Executive Order 11990, Protection of Wetlands
Floodplains:
• Is the project in a FEMA floodplain? ☒Yes ☐No ☐ Floodplain not mapped • ☒ Project is consistent with Executive Order 11988, Floodplain Management
Section 4(f) of the USDOT Act:
• Does the project meet the Section 4(f) exemption for Federal lands transportation facilities under Section 1119(c)(2) of MAP-21, 23 U.S.C. 138(a)? ☒Yes ☐No • Is there a use of a Section 4(f) property in the study area? ☐Yes ☒No If Yes, De Minimis Finding:☐ Programmatic:☐ Individual 4(f): ☐
Section 6(f) of the Land and Water Conservation Fund:
• Was the property purchased with grant funds from the Land and Water Conservation Fund?

☐Yes ☒No • If Yes, was documentation of approval from National Park Service Director received for the conversion or replacement of 6(f) property? ☐Yes ☐No
Coastal Zone Management Act of 1972:
• Is the project in a Coastal Zone? ☐Yes ☒No • If yes, will a Federal Consistency Review be completed? ☐Yes ☐No
Right of Way:
• Is the project completely within the transportation facility's right-of-way? ☒Yes ☐No • If no, will the project require relocations or easements? ☐Yes ☐No
Hazardous Waste and Materials:
• Are hazardous materials or contamination exceeding regulatory thresholds (as set by U.S. EPA, County Environmental Health, etc.) present? ☐Yes ☒No • If Yes, is the nature and extent of the hazardous materials or contamination fully known? ☐Yes ☐No, plan for securing information provided in Notes
Section 7(a) of the Wild and Scenic Rivers Act:
• Are there Wild and Scenic Rivers? ☐Yes ☒No ☐Eligible If Yes, has review by the river-administering agency been completed? ☐Yes ☐No
Clean Air Act:
• Is the project in a non-attainment area? ☒Yes[ii] ☐No If Yes, is the project on the Transportation Improvement Program(TIP)/State Implementation Plan(SIP)? ☐ Yes ☒No
Highway Traffic and Construction Noise Regulations:
• ☐ The proposed project is a Type I project (highway on a new location, substantial horizontal or vertical alteration, new through or auxiliary lanes). Noise analysis is required. • ☐ The proposed project is a Type II project (retrofit for noise abatement). • ☐ The proposed project is a Type III project (noise analysis not required). • ☒ Does not apply.
Environmental Justice and Title IV Act:
• Is an Environmental Justice population, as identified in Executive Order 12898, present? ☐Yes, explanation provided in Notes section ☒No • Will the project induce disproportionately high and adverse impacts to minority, low income, or special groups? ☐Yes, additional analysis is needed ☒No
Farmland Protection Policy Act for Highway Projects:
• Does the project displace, require acquisition of, or require an easement from farmland? ☐Yes ☒No
Greenhouse Gas Emissions and Climate Change

• Have greenhouse gas (GHG) emissions and climate change considerations been integrated into the planning, design, acquisition and/or construction processes? ☐Yes ☒No Comments: See below. • Have GHG emissions or reductions been quantified for the expected lifetime of the action? ☐Yes ☒No Comments: The proposed project involves repairs to pedestrian trails within the Wallkill River NWR, no permanent changes in GHG emissions would occur as a result of the proposed project. Temporary increases in GHG emissions would occur during project construction due to the presence of construction vehicles and equipment, but impacts will be minimized by implementing best management practices during project construction to reduce emissions generated from construction vehicles and equipment. • Does the project consider biogenic carbon dioxide sources, carbon stock, and the project's overall carbon footprint? ☐Yes ☒No Comments: N/A • Does the project include resilience and adaptation measures to address climate change? ☐Yes ☒No Comments: N/A
Does the project affect any other resources not listed above? Explain below and list anticipated permits:
Additional anticipated permits: New Jersey Department of Environmental Protection (DEP) Flood Hazard
Area Permit; NJDEP Freshwater Wetland GP; Upper Delaware Soil Conservation District Erosion and
Sediment Control (ESC) Plan review and approval.

[i] Federally-listed species per U.S. Fish and Wildlife Service's IPaC: Northern Long-Eared Bat (Myotis septentrionalis), Indiana Bat (Myotis sodalis), Bog Turtle (Glyptemys muhlenbergii), Monarch Butterfly (Danaus plexippus).

[ii] Sussex County, New Jersey is in non-attainment for the 2008 and 2015 8-hour ozone standard; however, a conformity determination is not required for the project as it is a listed action under 40 CFR 93.153 (c)(2)(iv). The project is also exempt from conformity analysis under 40 CFR 93.126 as it is an ERFO program project.

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U.S. Department of Transportation
Federal Highway Administration

Eastern Federal Lands Highway Division

22001 Loudoun County Parkway
Building E2, Suite 200
Ashburn, VA 20147

SENT VIA ELECTRONIC CORRESPONDENCE

In Reply Refer To: HFPP-15

Ms. Katherine Marcopul
Deputy State Historic Preservation Officer
New Jersey Department of Environmental Protection
Historic Preservation Office
P.O. Box 420
Trenton, NJ 08625-0420

Subject: NJ ERFO FW WARI 2019-1(1), Wallkill River National Wildlife Refuge Trail Repairs Section 106 Request for Concurrence

Dear Ms. Marcopul:

The Eastern Federal Lands Highway Division of the Federal Highway Administration (FHWA), in cooperation with the U.S. Fish and Wildlife Service (USFWS), is proposing to repair portions of the Liberty Loop and Dagmar Dale Nature trails within the Wallkill River National Wildlife Refuge (Refuge) located in Vernon Township, Sussex County, New Jersey (Figure 1) that were damaged by back-to-back storms in 2018. FHWA has been designated as the lead Federal agency for the undertaking for compliance with Section 106 of the National Historic Preservation Act.

Project Description

Proposed repairs on the Dagmar Dale Trail (Figure 2) will involve replacing failed culverts which cross under the trail and resurfacing and regrading damaged portions of the trail surface. The proposed repairs on the Liberty Loop Trail (Figure 3) will involve deep patch repairs of the trail embankments, installation of cellular confinement system (CCS), and turf re-establishment. Note that the scope of work on the Liberty Loop Trail provided in the enclosed preliminary plan set has changed. Due to New Jersey flood hazard area constraints, the trail surface elevation cannot be raised within the regulatory floodway; as such, the existing trail profile must be maintained. FHWA is proposing to install CCS along these portions of the trail to make the trail surface more resilient to floodwaters. This will require excavation of up to 10 inches of the existing trail surface to facilitate installation of the CCS. The pre-and post-construction trail elevation will be the same.

Identification Efforts

The project Area of Potential Effect (APE) has been defined as the existing trail surface and trail embankments within the project areas. Consultation with the Region 5 USFWS Regional Historic Preservation Officer (RHPO) found that several archaeological sites have been recorded within the Refuge and surrounding area, dating to the Middle Archaic to Late Woodland Periods.

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The distribution of archaeological sites within the area suggests a fairly intensive occupation of terraces within the Wallkill floodplain, especially at stream and river confluences.

Site 28-SX-273 (FWS Site No. WLK-004P) which was first identified in 1982, is a Late Archaic to Early Woodland archaeological site that was interpreted to be a processing area for Wallkill Valley chert. The site appears to overlap the Dagmar Dale Loop Trail west of Dagmar Dale Site 3. No other historic properties or archaeological sites are known to occur within or immediately adjacent to the Dagmar Dale Trail APE.

The Appalachian National Scenic Trail (A.T.) overlaps with the Liberty Loop Trail within the Wallkill National Wildlife Refuge for approximately 1 mile before turning onto Oil City Road near the New York-New Jersey boarder. A review of the New Jersey Department of Environmental Protection, Land Resource Protection Web Application on May 1, 2023, found that the portion of the A.T. that lies within the Liberty Loop Trail APE is eligible for listing in the National Register of Historic Places (SHPO Inventory ID 2778).

Determination of Effects

The proposed project would take place within the existing disturbed trail prisms and embankments. Although Site 28-SX-278 may overlap the Dagmar Dale Trail near Dagmar Dale Site 3, all proposed work would occur on the existing trail surface and within the existing trail prism. Therefore, FHWA has determined that the proposed undertaking would have **no adverse effect** on Site 28-SX-278.

Additionally, the proposed project would repair the existing Liberty Loop Trail surface and embankments and would not alter the surrounding viewshed for A.T. users. Therefore, FHWA has determined that the proposed undertaking would have **no adverse effect** on the Appalachian National Scenic Trail historic district.

A NJHPO Email Submittal Form is attached for your review. FHWA and USFWS respectfully requests your review and concurrence with our determinations above. Please provide us with a response within 30 days of this letter. If you require any additional information, or have any questions, please contact Ms. Kim Campo-Allen, Environmental Protection Specialist, at Kim.Campo-Allen@dot.gov or (703) 404-6374.

Sincerely yours,

KEVIN S ROSE

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Kevin Rose
Environmental Compliance Team Lead

Enclosures:
1. Figure 1, Location Map
2. Figure 2, Dagmar Dale Nature Trail Sites
3. Figure 3, Liberty Loop Trail Sites
4. Trail Sites Photo Sheet
5. NJHPO Email Submittal Form
6. Preliminary Engineering Plan Set

As proposed, the project will not adversely affect historic properties. Pursuant to 800.5(c), if no consulting parties object to this finding within the 30 day review period, the project may proceed, as proposed, unless resources are discovered during project implementation, pursuant to 800.13.

Katherine A. Marcopul 6/20/2023

KATHERINE A. MARCOPUL
Deputy State Historic Preservation Officer

Date

NAE

INTRA-SERVICE SECTION 7 BIOLOGICAL EVALUATION FORM

NJ ERFO WARI 2019-1(1)
Dagmar Dale Trail

Project Name: Improvements **Originating Person:** Anna Harris
Township: Sussex **Telephone Number:** 973-417-9542
County: Sussex, NJ **Date:** 6/1/2023
Project Code: 2023-0073676

I. Region: 5

II. Service Activity (Program): Wallkill River National Wildlife Refuge

III. Pertinent Species and Habitat:

A. Listed species and/or their critical habitat within the action area:
Indiana bat (*Myotis sodalis*), endangered
Northern long-eared Bat (*Myotis sodalis*), endangered
Bog turtle (*Glyptemys muhlenbergii*), threatened

B. Proposed species and/or proposed critical habitat within the action area:
Tricolored bat (*Perimyotis subflavus*), proposed endangered

C. Candidate species within the action area:
Monarch butterfly (*Danaus plexippus*)

D. Include species/habitat occurrences on a map.
See attached map depicting bat roost points, bat capture points, and documented bog turtle habitat around project area. Tricolored bat isn't shown on the map, but at least 4 individuals have also been captured during surveys within or immediately adjacent to the project area. Monarch butterflies have been observed in fields adjacent to the project area but are not likely to be present on the maintained trail or within forested sections of the project area.

Wallkill River NWR uses IPaC to identify threatened and endangered species, including for purposes of this Biological Evaluation. IPaC is occasionally updated to reflect the best available information on species ranges, so project species lists produced by IPaC are only valid for 90 days. Staff present on the refuge and conducting this evaluation may have the best available information about the presence of fish and wildlife species. Thus, where species are identified by IPaC, but the refuge has information that the species is not actually present within the "action area," we have explained that as the basis for our determination that any construction related activities will have no effect on the species.

IV. Description of proposed action (attach additional pages as needed):

This project is for trail improvements on the Dagmar Dale trail on the Wallkill River NWR. Proposed repairs on the Dagmar Dale Trail (Figure 1) will involve replacing failed culverts

which cross under the trail and resurfacing and regrading damaged portions of the trail surface. There are 6 sites identified for improvements. The construction of these trail resiliency projects is scheduled to begin in the Summer of 2024. All work will be in existing trail areas, making improvements to areas severely damaged by storms and impacts of climate change. The work will include replacement of culverts and rebuilding areas washed out by heavy rainfall. Construction vehicles will be in the area during the project, including excavators and backhoes.

Site 1: Culvert failure/blowout. Proposed work involves the removal of the existing failed 30" corrugated metal pipe culvert, installation of a 48" high density polyethylene (HDPE) pipe with headwalls, re-establishment of trail over culvert, and placement of class 4 riprap at culvert outlet to prevent scour. Trees within the limits of disturbance (5-6 in total) would be removed during the inactive season for the Indiana, northern long-eared bat, and tri-colored bat. A stream diversion would be required to facilitate project construction utilizing appropriate BMPs for site conditions including filter bags/dewatering devices.

Site 2: Trail soft spot. Proposed work consists of trail reconstruction with installation of approximately 200 linear feet of cellular confinement system (CCS).

Site 3: Trail soft spot with wetland draining across trail surface. Proposed work consists of the installation of a 24" elliptical reinforced concrete pipe (RCP) and reconditioning of approximately 200 linear feet of trail surface.

Site 4: Culvert failure. Existing failed culvert conveys stormwater across trail and discharges to Wallkill River tributary downstream of the project area. Proposed work consists of replacement of the existing 12" plastic pipe with at 24" RCP culvert.

Site 5: Trail soft spot. Proposed work consists of trail reconstruction with installation of approximately 400 linear feet of CCS.

Site 6: Culvert failure. Proposed work consists of the replacement of the existing damaged 24" by 42" concrete box culvert with two 48" HDPE pipes with headwalls. Class 4 riprap would be placed at the culvert outlet to prevent scour. A stream diversion would be required to facilitate project construction utilizing appropriate BMPs for site conditions including filter bags/dewatering devices. Due to access issues, riprap will likely be sourced from within the Refuge. Trees within the limits of disturbance (4-6 in total) would be removed during the inactive season for the Indiana, northern long-eared bat, and tri-colored bat.

V. Determination of effects:

A. Explanation of effects of the action on species and critical habitats in items III. A, B, and C (attach additional pages as needed):

Bog turtle – Bog turtles can be found in various locations throughout the Wallkill River valley in remnant open, wet bogs. A former emergent wetland located approximately 600 feet from Site 1 had been previously used by bog turtles, circa 2005. Radio telemetry data available for this bog turtle site indicates that tracked turtle(s) moved primarily between the emergent wetland and a forested wetland further south (approximately 1,500-1,600 ft from site 1) where the turtle(s)

hibernated. Off-site changes to hydrology rendered the former emergent wetland unsuitable for bog turtles, and no bog turtles have been observed in this portion of the refuge in recent years. The Wallkill Refuge considers the bog turtle one our focal species and we anticipate monitoring the construction area preconstruction to confirm our hypothesis that no bog turtles will be active or around the action area during the construction project.

The construction activity is taking place in the existing trail network on Dagmar Dale trail North and South. April 1-October 15 represent the period of activity for bog turtles, which overlaps with the proposed project work window. However, the radio tracked turtle(s) did not spend time in the wetlands associated with Site 1 or other sections of the Dagmar Dale trails. Bog turtles do occasionally make over land movements across the landscape, but the likelihood that they would be encountered on the existing trail network is low. If bog turtles were present on-trail, they could be killed or injured by vehicles or construction equipment moving between sites. The project also involves some work in wetlands and two small streams, which may contain suitable habitat for bog turtles. If bog turtles were present in any of the wetlands proposed for disturbance, bog turtles could potentially be killed or injured by the construction equipment and/or any potential bog turtle nests could be compromised. However, most wetlands within the project area lack characteristics of suitable bog turtle habitat (*e.g.*, deep mucky soils) and are unlikely to support the species.

Proposed project activities involve upgrading the size of one stormwater culvert at site 4 and two stream culverts at sites 1 and 6. These modifications to culvert size, as well as the temporary stream diversions needed for the culvert replacements at sites 1 and 6, may result in localized changes to wetland hydrology upgradient and downgradient of the culverts. The wetlands in the vicinity of sites 1, 4, or 6 are not likely to support bog turtles. However, the refuge anticipates monitoring the construction area preconstruction to confirm our hypothesis that no bog turtles occur around the action area.

Indiana bat, northern long-eared bat, and tricolored bat– Indiana bats were last documented on the refuge during surveys in summer of 2010. Northern long-eared bats were last documented on the refuge in summer 2013. Tricolored bats were last documented on the refuge in 2011. All three species are expected to be present within the project area during April 1-September 30. Indiana bats and northern long-eared bats typically roost in trees (particularly live trees and/or snags ≥3-5 inches dbh that have exfoliating bark, cracks, crevices, and/or hollows) during the day. Tricolored bats roost in live and dead leaf clusters of live or recently dead deciduous trees, typically in the mid to upper canopy area. During the active season, the bats emerge from roosts during the evening to forage in forested areas. After September 30 all three species are expected to migrate outside of the refuge, toward their fall swarming locations and hibernacula where individuals will spend the winter.

The construction activity on the Dagmar Dale trail is expected to occur throughout the summer and fall of 2024. Based on capture data from previous bat surveys, Indiana bats, northern long-eared bats, and tricolored bats likely use the Dagmar Dale trails and surrounding forest as commuting corridors and foraging habitat. Trees along the Dagmar Dale trails may also be suitable for roosting, but available radio telemetry data indicates that all tracked Indiana bats roosted in trees located at least 0.25 mile from the Dagmar Dale trail system. No northern long-

eared bats or tricolored bats captured along the Dagmar Dale trails were tracked to roosts. Construction activities may temporarily increase noise above baseline conditions in forested areas adjacent to the proposed project locations. However, potential disturbance to Indiana bats, northern long-eared bats, and tricolored bats is expected to be insignificant due to the relatively small scale of the proposed activities and temporary nature of the work. Construction activities will only occur during daytime hours, so project activities will not deter bats from foraging or commuting along the trails during the evening. The project also proposes removing a small number of trees at sites 1 and 6 (≤10 trees per location). Tree removal will be conducted only during the bats' inactive season (*i.e.*, October 1-March 31), when these species are not anticipated to be present at the refuge.

Monarch butterfly- Migratory generations of monarch butterflies commute through New Jersey during spring and fall migrations. Non-migratory generations of monarch butterflies breed in New Jersey during the summer months. The species relies on milkweed plants (*Asclepias spp.*) as the host plants for monarch caterpillars. Adult butterflies are generalists that can feed off of a wide variety of nectar-producing plants. Monarch butterflies require nectar plants with a diversity of flowering times, particularly fall blooming plants like goldenrod (*Solidago spp*.).

B. Explanation of actions to be implemented to reduce adverse effects:

Bog turtle:

- Refuge staff has conducted preliminary reconnaissance of the six sites for suitable bog turtle habitat and presence/absence of turtles by visually searching along the upstream and downstream vicinities of each site. Staff will continue monitoring during the active season until construction begins. If bog turtles are found near the project during monitoring, the conservation measures outlined will be revisited with the field office staff to determine any changes. No suitable habitat was found within the construction areas.
- To minimize the chance of injury to bog turtles and other native reptile and amphibian species that may be present, the Contractor will implement one of the below measures at sites where construction is proposed within or immediately adjacent to wetlands:
    - At the start of construction, a "herp" monitor, someone with experience monitoring native reptiles and amphibians, will survey the proposed work area for bog turtles. The herp monitor must then oversee the immediate installation of a turtle exclusion barrier (*e.g.*, straw bales, heavy duty silt fence, other approved barrier) that completely encompasses the work area; **or**
    - *At the start of each construction day*, the herp monitor will visually "sweep" work area for turtles immediately prior to the vehicle or equipment entering the area. The herp monitor will carefully check any areas with vegetation or deep muck that could obscure a turtle from view.
- The herp monitor will also survey trails or access roads for turtles ahead of vehicles when moving equipment between the work sites and/or staging areas.
- If a bog turtle is found, the herp monitor and a refuge biologist will determine whether the turtle can simply be moved out of the way (*e.g.*, if found on trail outside of work area), or if work needs to be paused until turtle exclusion measures (see above) can be installed to keep turtles out of the work area (*e.g*. if found in wetland in close proximity to where construction is occurring). The monitor must have the authority to pause construction activities.

Indiana bat, northern long-eared bat, and tricolored bat:

- Amount of tree removal has been minimized, and tree removal activities will be conducted only during October 1-March 31, when these species are not anticipated to be present at the refuge.

Monarch butterfly:

- If the work areas or access paths need to be larger than the existing trail, the Refuge will mow the work spaces ahead of the growing season when monarchs aren't present (*i.e.*, mow during October 1-May 1) or plan to survey for milkweed ahead of mowing activities and avoid any plants with monarch eggs/caterpillars on them.

The Service anticipates that the construction project is not likely adversely affect these species. Rather, we expect to see a positive habitat response by improving the trail network. By keeping visitors on the trail with the trail improvement projects, we anticipate less opportunities for the public to trample or otherwise disturb vegetation off-trail.

VI. Effect determination and response requested: [* = optional]

A. Listed species/designated critical habitat:

Determination	**Response requested**
no effect/no adverse modification (_____)	____ *Concurrence
may affect, but is not likely to adversely affect species/adversely modify critical habitat (species:__ Bog turtle, Indiana bat, Northern long-eared bat__)	__X_ Concurrence
may affect, and is likely to adversely affect species/adversely modify critical habitat (species:_____)	____ Formal Consultation

____Anna Harris_____ **___6/1/2023_____**
Project Biologist (Requestor) Date

VII. Reviewing ESFO Evaluation:

A. Concurrence____X_____ Nonconcurrence_____

B. Formal consultation required_____

C. Conference required_____

D. Informal conference required_____

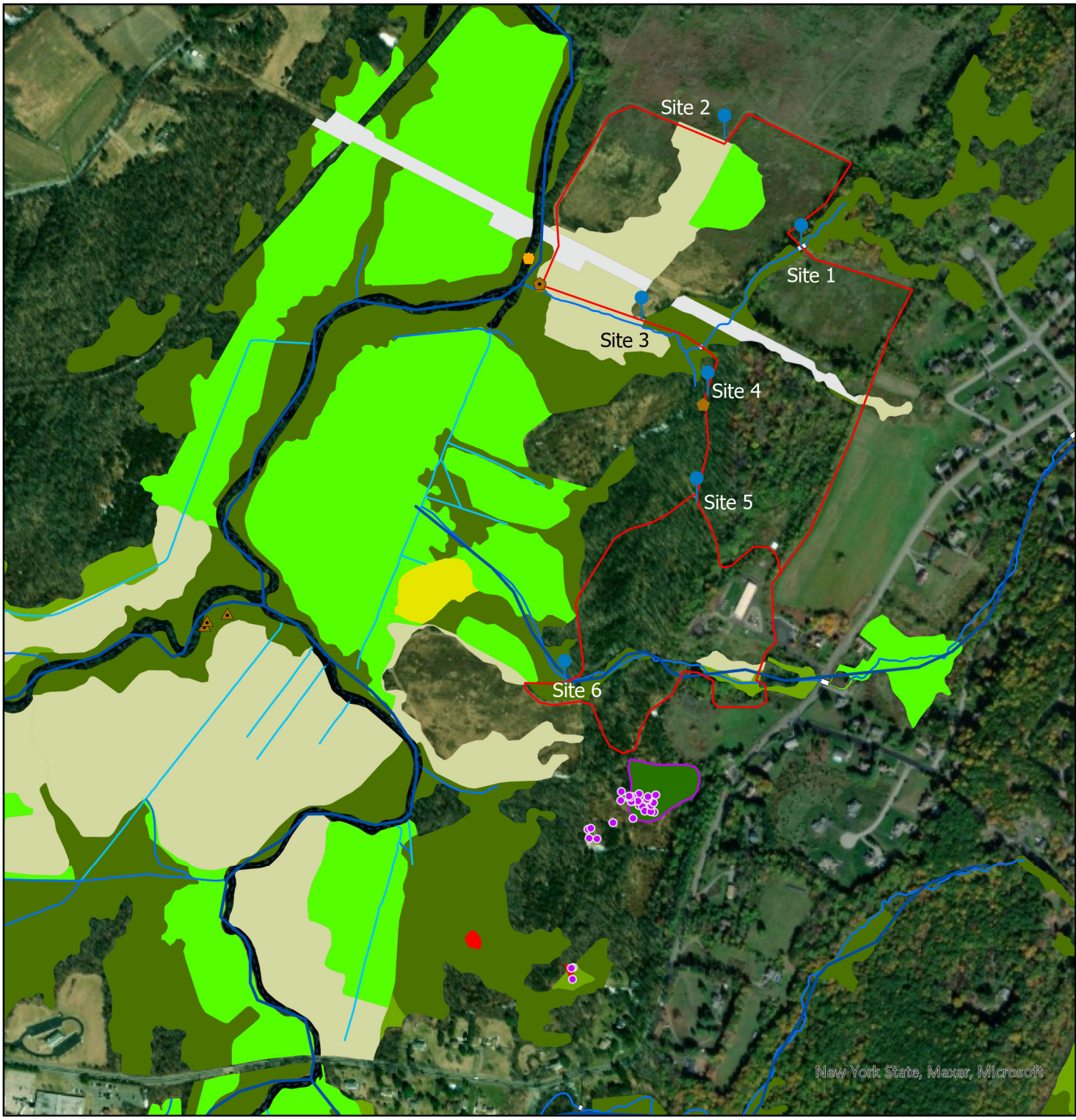
E. Remarks (attach additional pages as needed):

_____ _____
Endangered Species Biologist (Reviewer), New Jersey Field Office — Date

_____ _____
Supervisor, New Jersey Field Office — Date

Sensitive data- internal only

Wallkill NWR Dagmar Dale Trail- T/E Species Map



0 0.05 0.1 0.2 Miles

N

Indiana Bat Occurrence Points (Dec 2020)

- Capture
- Capture-maternity
- Maternity roost

NLEB OccurrencePoints (January 2022)

- Capture-maternity
- Bog turtle_2005_Features
- Bog turtle hibernacula
- Bog Turtle Polygons (2021)
- Major Rivers (2004)

Wetlands of New Jersey (2012)

- AGRICULTURAL WETLANDS (MODIFIED)
- FORMER AGRICULTURAL WETLAND (BECOMING SHRUBBY, NOT BUILT-UP)
- WETLAND RIGHTS-OF-WAY
- PHRAGMITES DOMINATE INTERIOR WETLANDS
- HERBACEOUS WETLANDS
- DECIDUOUS SCRUB/SHRUB WETLANDS
- DECIDUOUS WOODED WETLANDS
- MIXED WOODED WETLANDS (CONIFEROUS DOM.)

INTRA-SERVICE SECTION 7 BIOLOGICAL EVALUATION FORM

Project Name: FHWS ERFO Liberty Loop Trail **Originating Person:** Anna Harris
Township: Sussex, NJ and Pine Island, NY **Telephone Number:** 973-417-9542
County: Sussex, NJ and Orange County, NY **Date:** 6/1/2023

Distance to nearest town: 5 miles to nearest town of Sussex, NJ

I. Region: 5

II. Service Activity (Program): Wallkill River National Wildlife Refuge

III. Pertinent Species and Habitat:

A. Listed species and/or their critical habitat within the action area:

Indiana Bat (*Myotis sodalis*, endangered)
Northern Long-eared Bat (*Myotis septentrionalis*, endangered)
Eastern black rail (*Laterallus jamaicensis ssp. jamaicensis*, threatened)
Bog Turtle (*Glyptemys muhlenbergii*, threatened)

B. Proposed species and/or proposed critical habitat within the action area:

Tricolored Bat (*Perimyotis subflavus*, proposed endangered)

C. Candidate species within the action area:

Monarch Butterfly (*Danaus plexippus*, candidate)

D. Include species/habitat occurrences on a map.

Map(s) attached. See IPaC document.

IV. Description of proposed action (attach additional pages as needed):

This project is for trail improvements on the Liberty Loop Trail (Figure 1) and construction will involve deep patch repairs of the trail embankments, installation of cellular confinement system (CCS), and turf re-establishment. There will be an excavation of 10 inches of the existing trail surface to facilitate installation of the CCS.

The pre-and post-construction trail elevation will be the same. The construction equipment will be staged offsite and will use existing parking areas on the trail system. All construction and trail maintenance will occur during daylight hours. There is no tree clearing associated with this trail resiliency project.

V. Determination of effects:

A. Explanation of effects of the action on species and critical habitats in items III. A, B, and C (attach additional pages as needed):

Bog turtle – The project action area is approximately 2.5 miles from the nearest bog turtle occurrence. The project area consists of modified wetlands, which are not preferred by bog turtle. The surrounding wetlands are primarily forested with small patches of emergent wetland that will not be impacted by the proposed project activities. Wallkill Refuge frequently surveys for bog turtles and has no record of a bog turtle on the Liberty Loop trail or surrounding areas. Wallkill Refuge anticipates monitoring the construction area preconstruction to confirm no bog turtles will be present in or surrounding the action area during the construction project. Based on the lack of suitable bog turtle within the project action area and anticipated monitoring, no adverse effects are anticipated for bog turtle.

Indiana bat, northern long-eared bat, and tricolored bat- The proposed project does not require tree removal; therefore, no effects to Indiana bat, northern long-eared bat, or tricolored bat summer roosting habitat or commuting areas are anticipated.

Eastern black rail- Eastern black rail occurrences have been documented within close proximity to the Liberty Loop trail. Vegetation within the project limits of disturbance consists primarily of upland species that do not provide preferred nesting habitat for the eastern black rail. Additionally, repairs to the Liberty Loop Trail will not require the placement of fill in wetlands and no equipment would be allowed to enter wetlands. All work would occur on the existing trail surface and embankment. No changes in wetland hydrology would result from the trail repairs as no changes in trail surface elevation would occur and the project does not involve work to the existing water control structures at Liberty Loop. Additionally, temporary erosion and sediment (E&S) control devices would be installed prior to ground disturbing activities. E&S measures would include installation of silt fence and watering of exposed soils to prevent dust from entering adjacent wetlands. Mitigation measures to prevent the spread of invasive species within the Refuge would include inspection of construction equipment prior to entry into the project area. Vehicles and equipment would be inspected for any vegetation and/or vegetation parts (seeds, stems, etc.) to avoid introduction of invasive species. Based on the lack of disturbance to wetlands and the implementation of E&S and mitigation measures, no adverse effects are anticipated for eastern black rail.

Construction on the trail improvement project will occur during the summer and fall of 2024. Given the narrow width of the Liberty Loop trail system, the construction

equipment will be smaller than most traditional equipment. This will lessen the impacts from noise during the daylight hours for any nesting or migratory eastern Black rail.

Monarch butterfly- No mowing/removal of milkweed is proposed. Based on lack of impacts to suitable monarch butterfly habitat, no effects to monarch butterfly are anticipated.

B. Explanation of actions to be implemented to reduce adverse effects:

The Service anticipates that the construction project program will not likely adversely affect or have no effect on these species. Rather, Wallkill Refuge expects to see a positive habitat response by improving the trail network. This portion of the refuge has been severely damaged due to high water events during storms or hurricane events. The construction project will create a more climate resilient trail system at the Liberty Loop location.

As mentioned previously, Wallkill Refuge anticipates monitoring the construction area preconstruction to confirm no bog turtles will be present in or surrounding the action area during the construction project. If bog turtle is found before or during construction, either the New Jersey Field Office or the New Jersey Department of Environmental Protection will be contacted promptly for further guidance.

To reduce potential adverse effects to eastern black rail, a qualified monitor will perform routine inspection to monitor that erosion and sediment controls are functioning properly, invasive controls are being implemented, and there are no intrusions into the modified wetlands. The contractor will also reduce noise emissions as much as practicable, including use of modern equipment with better engine insulation and mufflers.

VI. Effect determination and response requested: [* = optional]

A. Listed species/designated critical habitat:

Determination	**Response requested**
no effect/no adverse modification (species: Indiana bat, northern long-eared bat)	__X_ *Concurrence
may affect, but is not likely to adversely affect species/adversely modify critical habitat (species: bog turtle, eastern black rail)	__X_ *Concurrence
may affect, and is likely to adversely affect species/adversely modify critical habitat	

(species:_____) ____ Formal Consultation

B. Proposed species/proposed critical habitat:

Determination	**Response requested**
No effect on proposed species/no adverse modification of proposed critical habitat (species: tricolored bat)	__X__ *Concurrence
Is likely to jeopardize proposed species/ adversely modify proposed critical habitat (species: None)	_____ Conference

C. Candidate species

Determination	**Response requested**
No effect on candidate species (species: monarch butterfly)	__X__ *Concurrence
Is likely to jeopardize candidate species (species:_____)	_____ Conference

____Anna Harris_____ **___6/1/2023_____**
Project Biologist (Requestor) Date

VII. Reviewing ESFO Evaluation:

A. Concurrence_____X_____ Nonconcurrence_____

B. Formal consultation required_____

C. Conference required_____

D. Informal conference required_____

E. Remarks (attach additional pages as needed):

_____ _____
Endangered Species Biologist (Reviewer), Date
New Jersey Field Office

_____ _____
Supervisor, New Jersey Field Office Date